

ABC SENIOR SECONDARY SCHOOL

CBSE CLASS XII — BOARD EXAMINATION (THEORY)

Subject: Science / Mathematics

Time Allowed: 3 Hours | Maximum Marks: 80 | Date: _____

Name: _____ Roll No.: _____

Class/Section: _____

GENERAL INSTRUCTIONS

- 1. All questions are compulsory.*
- 2. Read each question carefully before answering.*
- 3. Figures to the right indicate full marks.*
- 4. Use of calculator is not permitted unless stated.*

SECTION A — MULTIPLE CHOICE QUESTIONS

Each question carries 1 mark. Select the correct option.

Q1. (1 marks)

Which of the following statements about fuses is correct according to the given context?

- (a) Fuses should be replaced with wires or metal strips when they blow.
- (b) Fuses without the ISI mark can be used in specific applications.
- (c) Proper fuses with the ISI mark should be used for specified applications.
- (d) Fuses are not necessary for electrical safety.

Q2. (1 marks)

What type of image can be formed by a concave mirror according to the given context?

- (a) Enlarged and erect
- (b) Real, enlarged and inverted
- (c) Virtual and inverted
- (d) Cannot form a real image on a screen

Q3. (1 marks)

A circuit breaker (MCB) is used to protect electrical circuits. If a circuit is overloaded, the MCB will trip and cut off the power supply. Consider the following statements about MCBs and select the correct one: (A) MCBs are designed to trip only when the current exceeds a certain threshold. (B) MCBs can be used interchangeably with any type of fuse. (C) MCBs are not required in circuits with low current. (D) MCBs should

always be replaced with the same type of MCB. ?

- (a) (A) and (D)
- (b) (B) and (C)
- (c) (A) and (B)
- (d) (A) and (C)

Study the figure below.

Q4. (1 marks)

Boojho is experiencing a discrepancy in the temperature perception of the same water in mug C. What should he conclude from this observation?

- (a) The water in mug C is at a constant temperature.
- (b) There might be an issue with his sensory perception.
- (c) The mugs are not of the same material.
- (d) The water in mug C has changed temperature.

Q5. (1 marks)

What is the term used to describe the combination of two or more cells?

- (a) Battery
- (b) Cell
- (c) Generator
- (d) Transformer

Q6. (1 marks)

What happens when current is switched 'on' in a room heater?

- (a) The heater cools down.

- (b) The heater heats up.
- (c) The heater remains at room temperature.
- (d) The heater becomes a conductor.

Q7. (1 marks)

An electrician explained that fuses are used to protect electrical circuits from damage. Which of the following best describes the function of a fuse in an electrical circuit?

- (a) A To increase the current flow
- (b) B To protect the circuit from overcurrent
- (c) C To regulate the voltage
- (d) D To store electrical energy

Study the figure below.

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Q8. (1 marks)

When comparing electromagnets with different numbers of turns, what observation can be made about the relationship between the number of turns and the strength of the electromagnet?

- (a) A The strength decreases as the number of turns increases
- (b) B The strength increases as the number of turns increases
- (c) C The strength is unrelated to the number of turns
- (d) D The strength decreases as the number of turns decreases

SECTION B — VERY SHORT ANSWER

Answer in 40-60 words. Each question carries 2 marks.

Q9. (2 marks)

Refer to the distance-time graph in the diagram to determine which vehicle is moving faster. What does a steeper slope on the graph indicate about the vehicle's speed? (a) Vehicle A (b) Vehicle B

Figure



Q10. (2 marks)

Identify the distance-time graph that shows a truck moving with a constant speed. What characteristic of the graph indicates a constant speed? (a) Graph X (b) Graph Y

Figure



SECTION C — SHORT ANSWER

Answer in 60-80 words. Each question carries 3 marks.

Q11. (3 marks)

Boojho experiences a conflict in his perception of the temperature of water in mug C. After placing his hands in mugs A and B for 2-3 minutes, he simultaneously places them in mug C. What might be the reason for the difference in his initial perception and the final sensation in mug C? (3 marks)

Q12. (3 marks)

What is the term used to describe the combination of two or more electric cells? (3 marks)

SECTION D — CASE BASED / SOURCE BASED

Read the passage and answer the sub-questions. Each question carries 4 marks.

Q13. (4 marks)

Consider the following statements about indicators used in chemistry to determine the nature of a solution. Determine the correctness of the statements provided.

(a) Which of the statements (a), (b), (c), and (d) are correct?

(b) Explain why the correct statements are true in the context of indicator behavior?

SECTION E — LONG ANSWER

Answer in detail with diagrams where necessary.

Q14. (5 marks)

Refer to the diagram in Fig_0001, which illustrates the operation of an electromagnet. Boojho and Paheli are curious about the functioning of the circuit and the properties of

materials. Explain how the circuit in the diagram works to produce sound, and discuss why caterpillars shed their skin as they grow, contrasting this with human development. (a) Describe the sequence of events that occur when the iron strip is in contact with the contact screw and the hammer strikes the gong. (b) Explain the biological reason for caterpillars shedding their skin and why humans do not experience a similar process. ?

Figure

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SECTION F — EXTENDED RESPONSE

Answer with proper derivation / explanation. (8 marks)

Q15. (8 marks)

Refer to the diagram provided in Fig_0001, which illustrates the distance-time graph for an object moving with a constant speed. Boojho and Paheli are curious about the properties of this graph and the concept of constant speed. Using the information from the lesson and the diagram, explain why the distance-time graph for an object moving with a constant speed is a straight line. Additionally, discuss how the slope of this line relates to the speed of the object. (8 marks)

Figure

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